

Conjugation is not independent data: a formalized obstruction for Diaz's modulus question

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Abstract

Let $K \subseteq \mathbb{C}$ be a conjugation-stable subfield and let $u \in \mathbb{C}$ satisfy $u\bar{u} \in K$. We record an elementary observation and its consequences, all machine-checked in Lean 4 with Mathlib: since $\bar{u} = (u\bar{u})/u$, complex conjugation on the subfield generated by K and u is determined by the ring structure rather than being extra data. Three consequences follow. Any ring homomorphism fixing K and sending u to a second point t on the same circle automatically intertwines conjugation on that subfield; such a homomorphism exists on the subfield whenever u and t are transcendental over K ; and no vanishing-coefficient statement over K built from the associated 2×2 matrix distinguishes u from t . These are negative results about a class of proof strategies. No new transcendence result is proved, and Diaz's question remains open.

1 Setting

Write $\overline{\mathbb{Q}}$ for the field of algebraic numbers in $\mathbb{C}$ and

$$\mathcal{L} = \{u \in \mathbb{C} : e^u \in \overline{\mathbb{Q}}^\times\}$$

for the logarithms of algebraic numbers. Diaz [1] studied the use of complex conjugation in transcendence arguments for the exponential function, asking in particular how the non-holomorphic maps $z \mapsto \bar{z}$ and $z \mapsto |z|$ could enter such a proof at all.

Remark (to be completed). The precise statement and location of the modulus question — that no non-zero element of $\mathcal{L}$ has algebraic modulus — must be quoted from [1] with its section or page number before this note is circulated. Nothing below depends on the attribution: the results are proved for an arbitrary conjugation-stable K and are stated that way.

Throughout, $K \subseteq \mathbb{C}$ is a subfield with $\bar{a} \in K$ for every $a \in K$. The case of interest is $K = \overline{\mathbb{Q}}$.

Definition 1.1. For K a subfield of $\mathbb{C}$ and $u \in \mathbb{C}$, let $\text{hull}_K(u)$ denote the subfield of $\mathbb{C}$ generated by K together with u .

Call u a *candidate over K* if $u\bar{u} \in K$; the intended case is $u \in \mathcal{L}$ with $|u|$ algebraic, so that $u\bar{u} = |u|^2 \in \overline{\mathbb{Q}}$.

2 The mechanism

The whole note rests on one identity.

Lemma 2.1. *If $u \neq 0$ and $\rho = u\bar{u}$, then $\bar{u} = \rho/u$.*

Proof. Immediate from $\rho = u\bar{u}$ and $u \neq 0$. □

Trivial as it is, this is the point. For a candidate, ρ lies in K , so $\bar{u}$ is a rational function of u with coefficients in K . Conjugation on $\text{hull}_K(u)$ is therefore not additional structure that a homomorphism might fail to respect; it is already fixed by the field operations.

3 Rigidity

Lemma 3.1. *Let $f, g: \mathbb{C} \rightarrow \mathbb{C}$ be ring homomorphisms agreeing on K and at u . Then $f = g$ on $\text{hull}_K(u)$.*

Theorem 3.2. *Let $u, t \neq 0$ with $u\bar{u} \in K$ and $t\bar{t} = u\bar{u}$. If $\Phi: \mathbb{C} \rightarrow \mathbb{C}$ is a ring homomorphism fixing K pointwise with $\Phi(u) = t$, then*

$$\Phi(\bar{z}) = \overline{\Phi(z)} \quad \text{for every } z \in \text{hull}_K(u).$$

Proof. Both $\Phi \circ \text{conj}$ and $\text{conj} \circ \Phi$ are ring homomorphisms agreeing on K , because K is conjugation-stable and Φ fixes it. At u they agree too: by Lemma 2.1, $\Phi(\bar{u}) = \Phi(\rho/u) = \rho/t$ and $\bar{t} = \rho/t$. Now apply Lemma 3.1. $\square$

The intertwining is not arranged. It is forced.

4 Existence

Theorem 3.2 is conditional on a homomorphism existing. On the hull, one does.

Theorem 4.1. *If u and t are both transcendental over K , there is a K -algebra homomorphism $\Phi: K(u) \rightarrow \mathbb{C}$ with $\Phi(u) = t$.*

Proof. For f transcendental over K , the field $K(f)$ is K -isomorphic to the field of rational functions $K(X)$, by $X \mapsto f$. Applying this to u and to t gives $K(u) \cong K(X) \cong K(t)$ over K ; compose with the inclusion $K(t) \hookrightarrow \mathbb{C}$. $\square$

Remark (scope of Theorem 4.1). The homomorphism is produced on $K(u)$, not on all of $\mathbb{C}$. Extending a homomorphism from $K(u)$ to an endomorphism of $\mathbb{C}$ is possible, by extending a transcendence basis and using that $\mathbb{C}$ is algebraically closed, but it is not proved here and is not needed: every use below evaluates Φ only on $\text{hull}_K(u)$.

5 The obstruction

Attach to a candidate the matrix

$$H(u, r) = \begin{pmatrix} u & r \\ r & \bar{u} \end{pmatrix},$$

where $r \in K$ satisfies $u\bar{u} = r^2$. Then $\det H = u\bar{u} - r^2 = 0$, so H has rank one, and its coefficients factor.

Theorem 5.1. *Let $r \in K$, let $u \notin K$ with $u\bar{u} = r^2$, and let $w, v \in K^2$ be non-zero. Then*

$$\sum_{i,j} w_i H(u, r)_{ij} v_j \neq 0.$$

Corollary 5.2. *Let $r \in K$ and let $u, t \notin K$ both satisfy $u\bar{u} = t\bar{t} = r^2$. Then for all non-zero $w, v \in K^2$,*

$$\sum_{i,j} w_i H(u, r)_{ij} v_j = 0 \iff \sum_{i,j} w_i H(t, r)_{ij} v_j = 0.$$

Proof. By Theorem 5.1 neither side ever holds. $\square$

Corollary 5.2 is the negative statement. A proof strategy that accumulates vanishing-coefficient constraints over K on u and $\bar{u}$, hoping they collide, cannot separate a candidate from an ordinary point placed on the same circle. The reason is Lemma 2.1: the constraints are not independent, because $\bar{u}$ was never independent of u .

6 What is and is not claimed

Claimed: the statements above, for an arbitrary conjugation-stable subfield K , each machine-checked.

Not claimed. No transcendence result, and no progress on whether a non-zero element of $\mathcal{L}$ can have algebraic modulus; that question is untouched and remains open. Nothing here shows the conjugation method fails in general — only that this particular family of statements, vanishing of coefficients over K of the rank-one matrix H , does not separate candidates. Theorem 4.1 is proved on $K(u)$ only. The formalization does not assert that any candidate exists; establishing that a non-zero $u \in \mathcal{L}$ with $|u|$ algebraic is transcendental requires Hermite–Lindemann, which the compared surface of this note deliberately avoids.

7 Related work and search

Remark (to be completed by the author). This section must record an explicit literature search: which databases and review services were consulted, with what queries and on what date; which adjacent results were examined; and what was concluded about whether the observations above are new. If the search does not settle novelty, this section must say that novelty is unknown rather than assert priority. Author’s existing search notes and references are to be inserted here.

8 Formalization

The development is Lean 4 with Mathlib; sources at <https://github.com/carlok/diaz-modulus-lean>. The advertised statements are in `Challenge.lean`, with proofs in `Solution.lean`.

Statement	Lean declaration	Module
Lemma 2.1	<code>conj_eq_rho_div</code>	<code>Diaz.Closure</code>
Lemma 3.1	<code>eq0n_hull</code>	<code>Diaz.Closure</code>
Theorem 3.2	<code>conj_comm</code>	<code>Diaz.Transfer</code>
Theorem 4.1	<code>exists_algHom_of_transcendental</code>	<code>Diaz.Palomar</code>
Theorem 5.1	<code>no_vanishing_coeff_matrix</code>	<code>Diaz.Rigidity</code>
Corollary 5.2	<code>coeff_indistinguishable</code>	<code>Diaz.Palomar</code>
$\text{hull}_K(u)$	<code>hull</code>	<code>Diaz.Closure</code>
$H(u, r)$	<code>Hmat</code>	<code>Diaz.Rigidity</code>

Running `#print axioms` on each of these lists only `propext`, `Classical.choice` and `Quot.sound`. The wider repository also contains two imported transcendence statements declared as axioms, `hermite_lindemann` and `exists_ringHom_of_transcendental`, used for results outside the list above. No statement in the table depends on either.

References

- [1] G. Diaz, *Utilisation de la conjugaison complexe dans l’étude de la transcendance de valeurs de la fonction exponentielle usuelle*, J. Théor. Nombres Bordeaux **16** (2004), no. 3, 535–553, [doi:10.5802/jtnb.459](https://doi.org/10.5802/jtnb.459).
- [2] The Mathlib Community, *Mathlib: the mathematical library of Lean 4*, [project repository](https://project-repository.org).